

Special Seminar on Multi-Modal AI

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<https://cis7000-fall26.thelabone.org/>

Lecture 5: Vision-Language Learning

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Initial Proposal

Deadline: This Sunday! Each project has ≤ 4 students

This is initial proposal: I will provide feedbacks

Proposal Structure: Science

- What is a hypothesis you want to test?
- What are existing studies?
- What are the inputs, outputs, datasets, objectives, and metrics?
- What are the control experiments?
- What are possible outcomes? And what are the contributions to the community?

Proposal Structure: Application

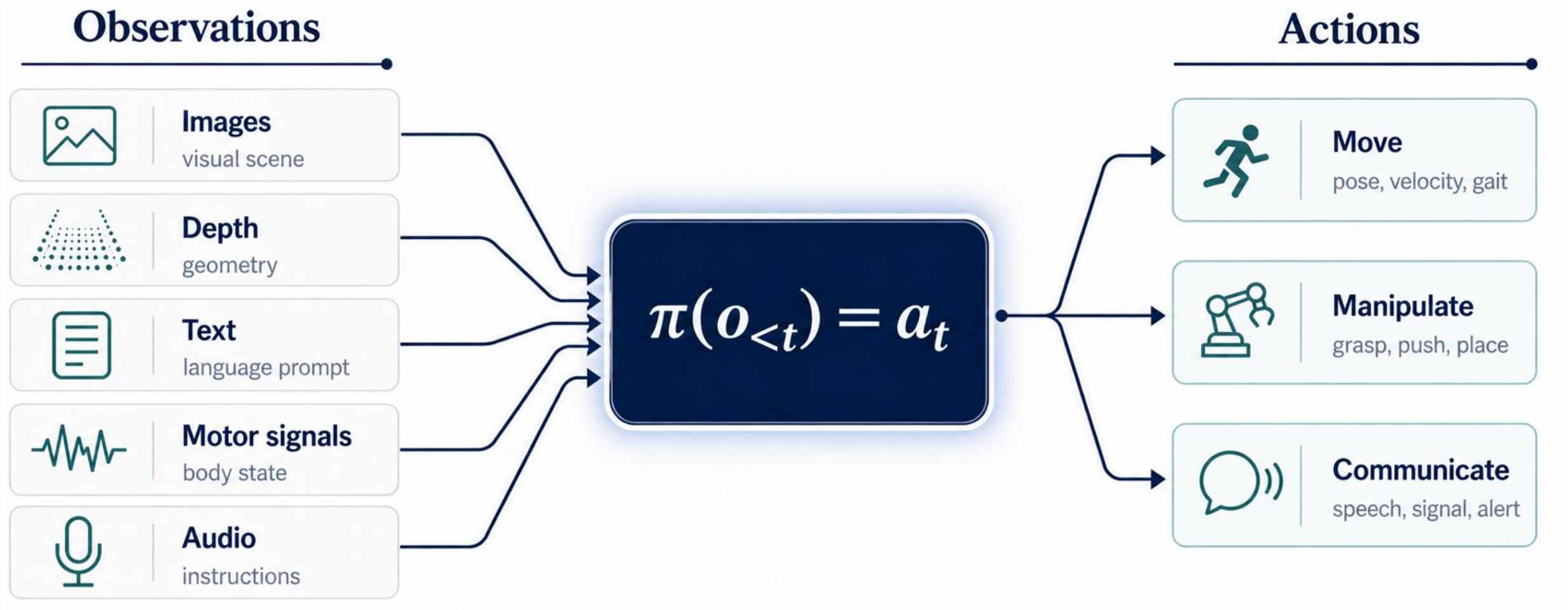
- What is the problem you want to solve?
- What are existing studies?
- What are the inputs, outputs, datasets, objectives, and metrics?
- What are the baselines?
- What are the limitations or future work you may encounter?

Goal of This Course

Use Machine Learning to Build Intelligent Agents

Abstract View of An Agent: Multimodal AI

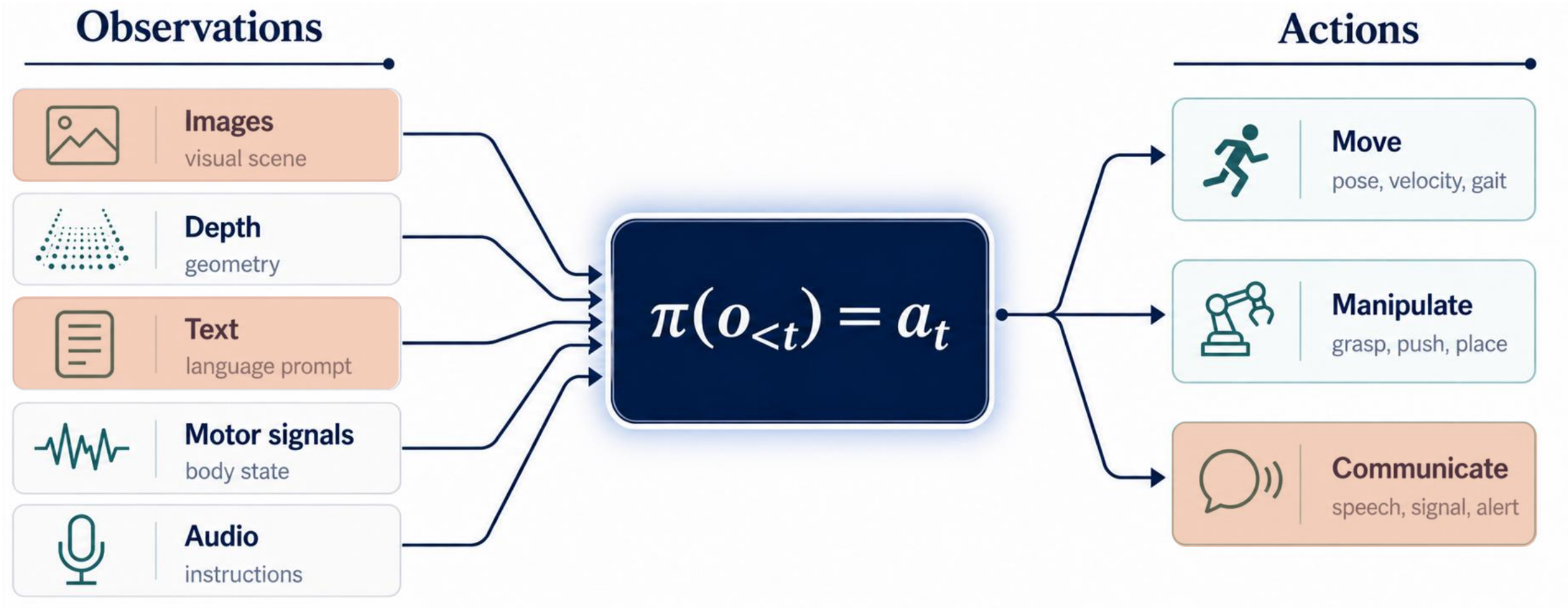
π : observations $\rightarrow$ action



Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Abstract View of An Agent: Multimodal AI

π : observations $\rightarrow$ action



Notation: t – time. $o_{<t}$ - all observations before t . a_t - action at t .

Goal of This Lecture

Use machine learning to connect vision and language

Today's Agenda

Part 1: Principles for Vision-Language Alignment

- Contrastive learning and embedding geometry

Part 2: Principles for Vision-Language Alignment Structures

- Scene graphs and compositional structures

Today's Agenda

Part 1: Principles for Vision-Language Alignment

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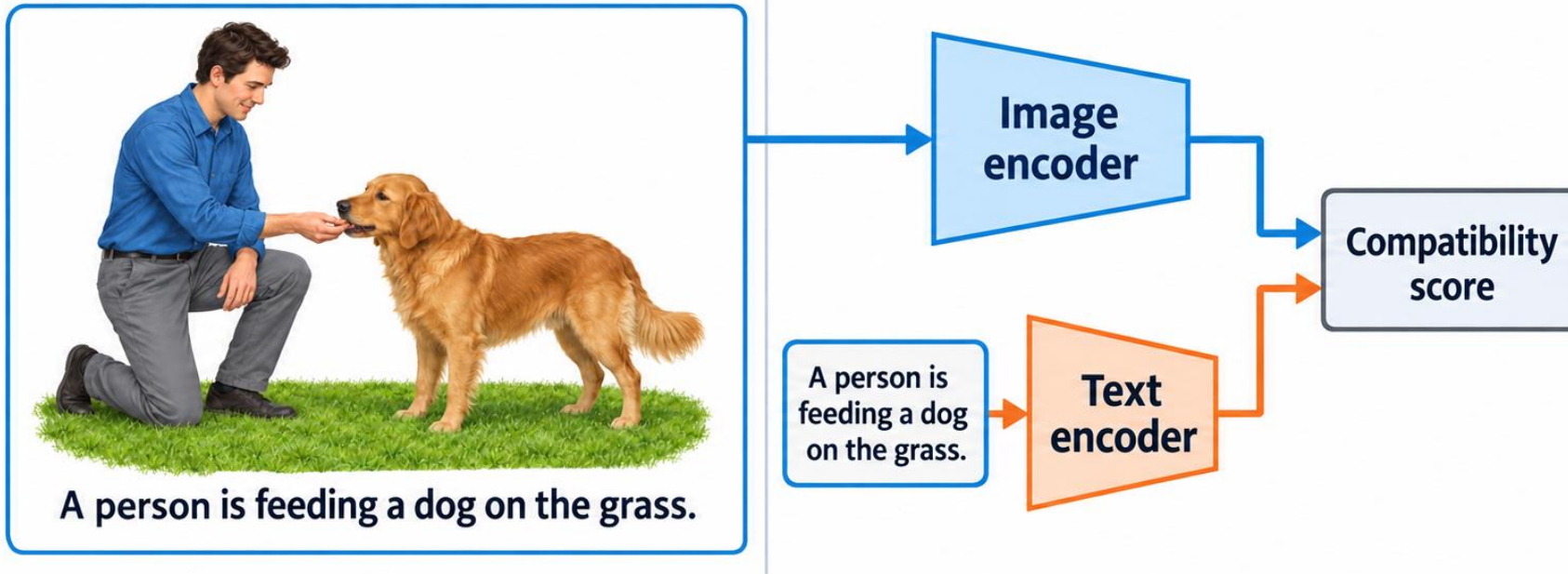
Vision-Language Alignment

Training Pairs



Vision-Language Alignment

Training Pairs



Vision-Language Alignment

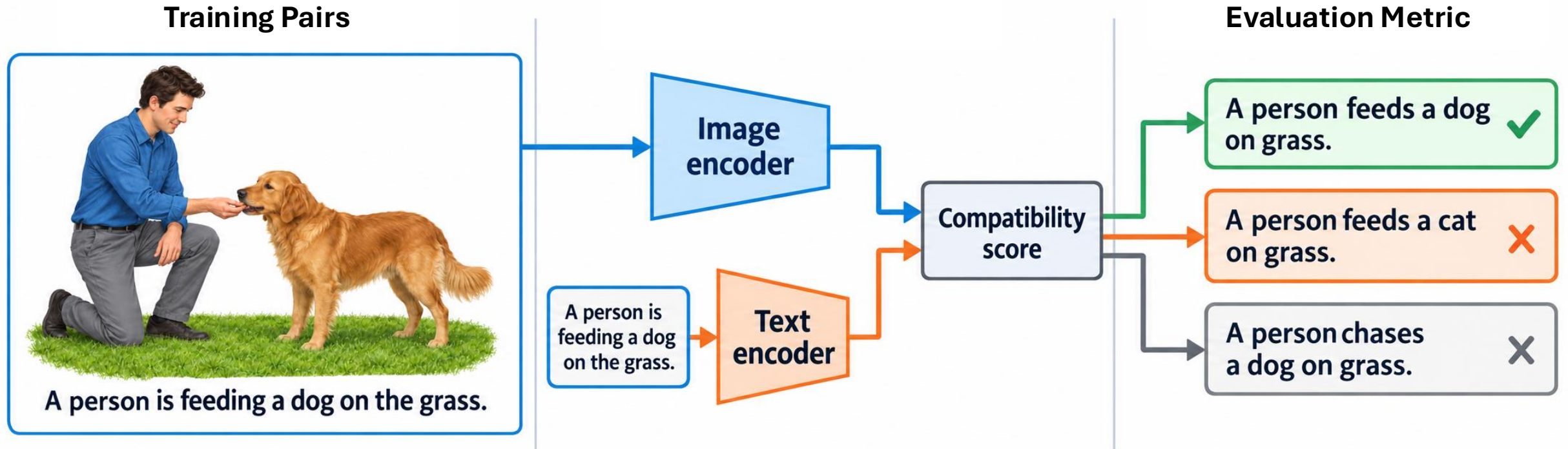
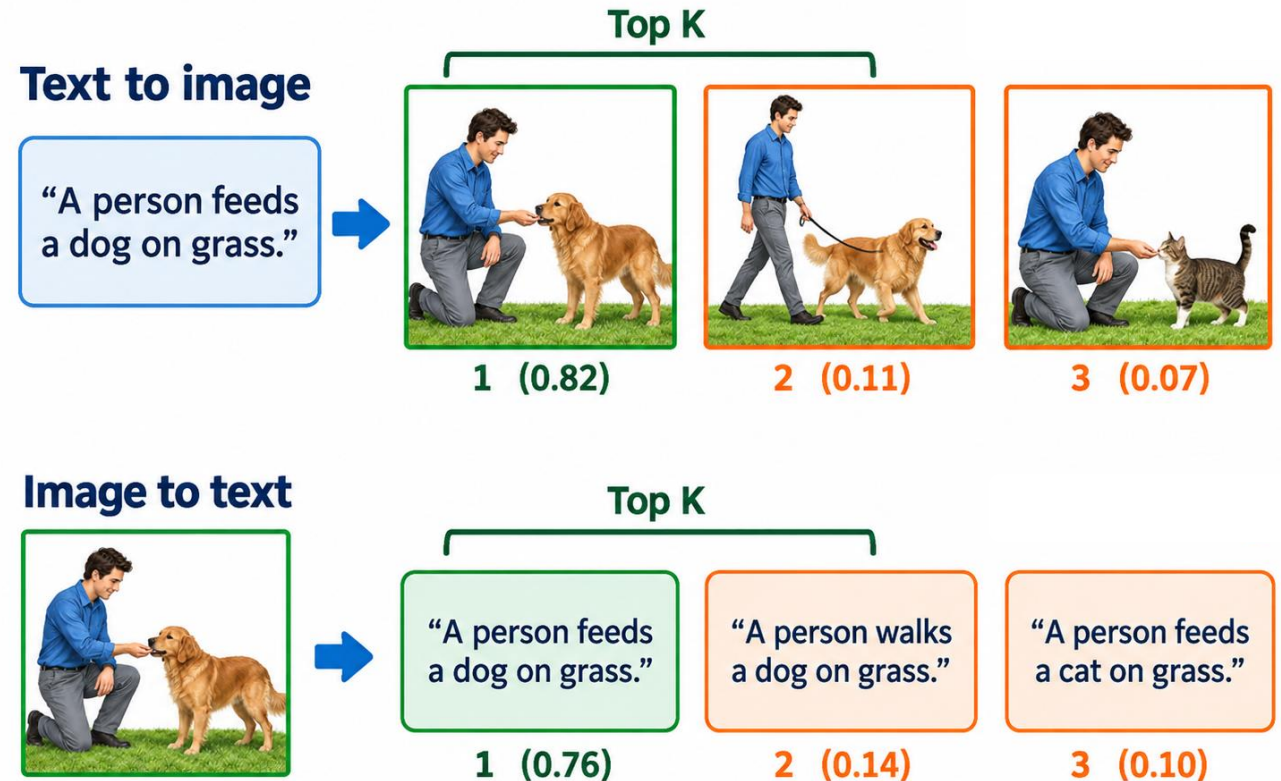


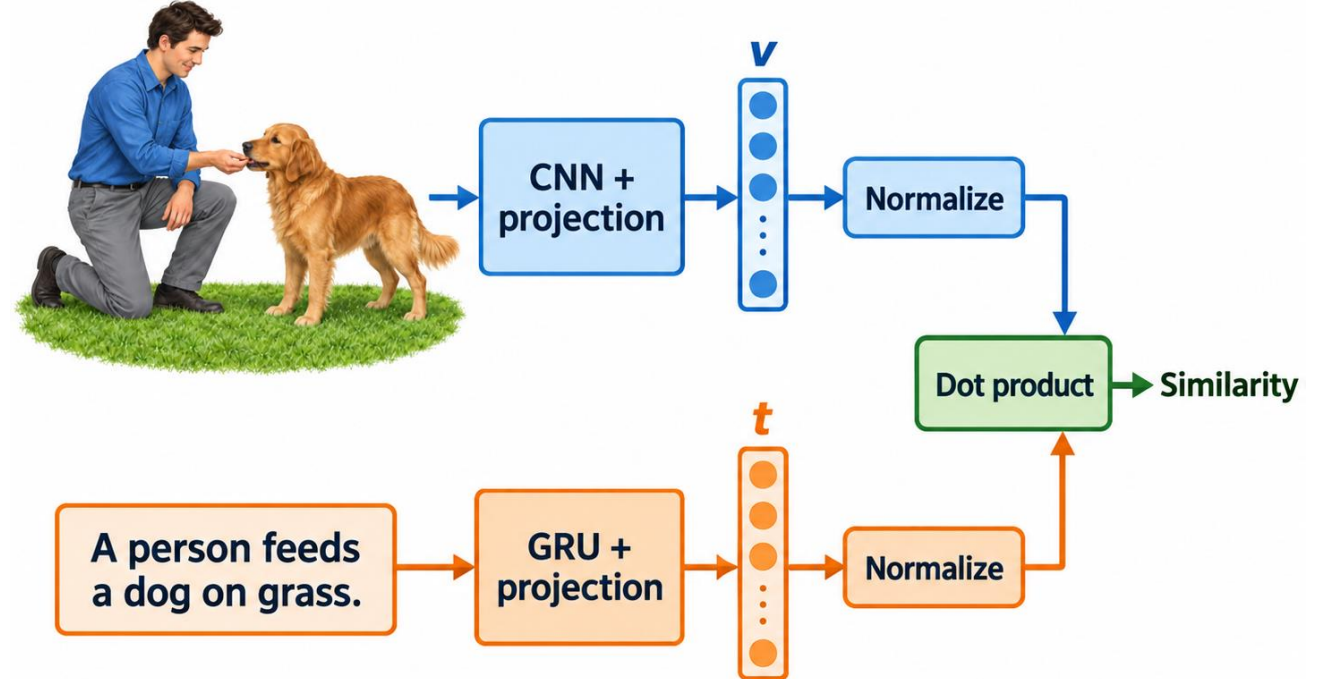
Image-Text Retrieval

- Text query: rank images by relevance
- Image query: rank captions by relevance
- Recall@K: fraction of queries with a correct item in the top K



VSE: Basic Architecture for Alignment

- CNN features and a learned projection encode images
- A GRU and projection encode captions
- Normalize both vectors and compare their dot product



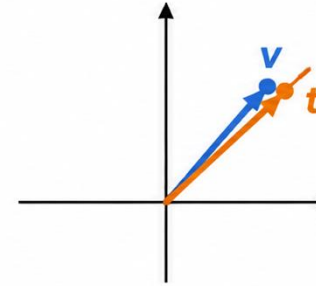
Visual-Semantic Similarity

$$v = \text{normalize}(W_v^T \varphi(I))$$

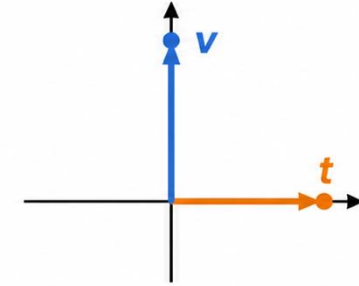
$$t = \text{normalize}(W_t^T \psi(T))$$

$$s(I, T) = v^T t$$

Unit-length vectors



Higher similarity



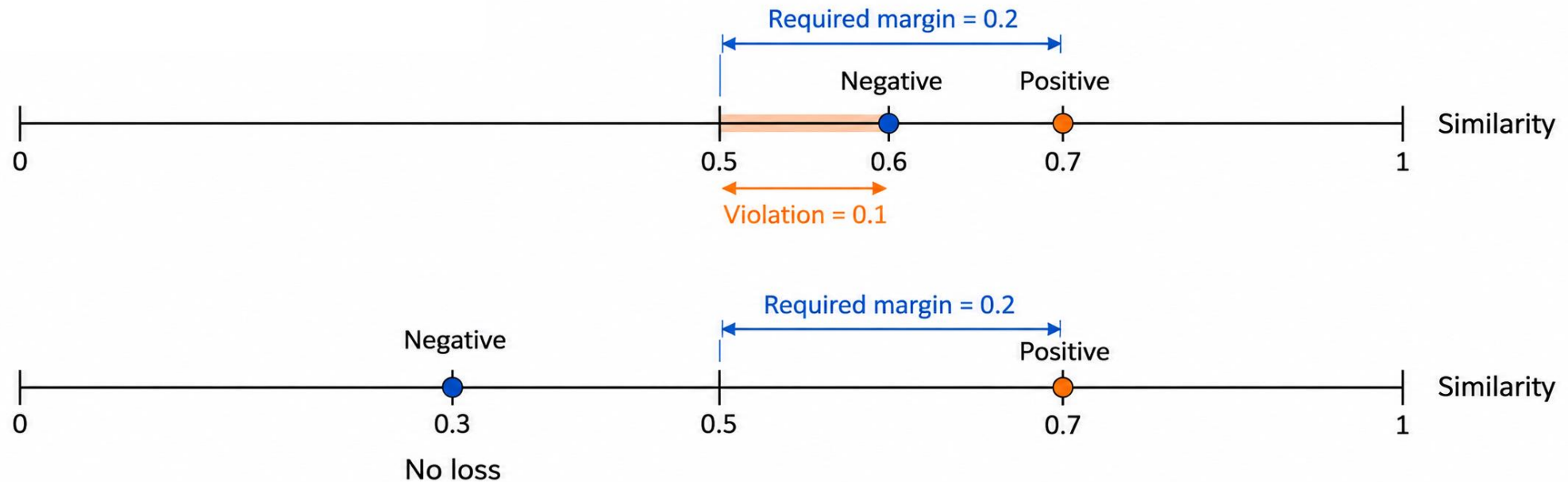
Lower similarity

Ranking Loss

Positive pair: (I, T) Negative caption: T'

$$\ell(I, T, T') = \max(0, \alpha - s(I, T) + s(I, T'))$$

Issue: R@1 fails as soon as one negative ranks first

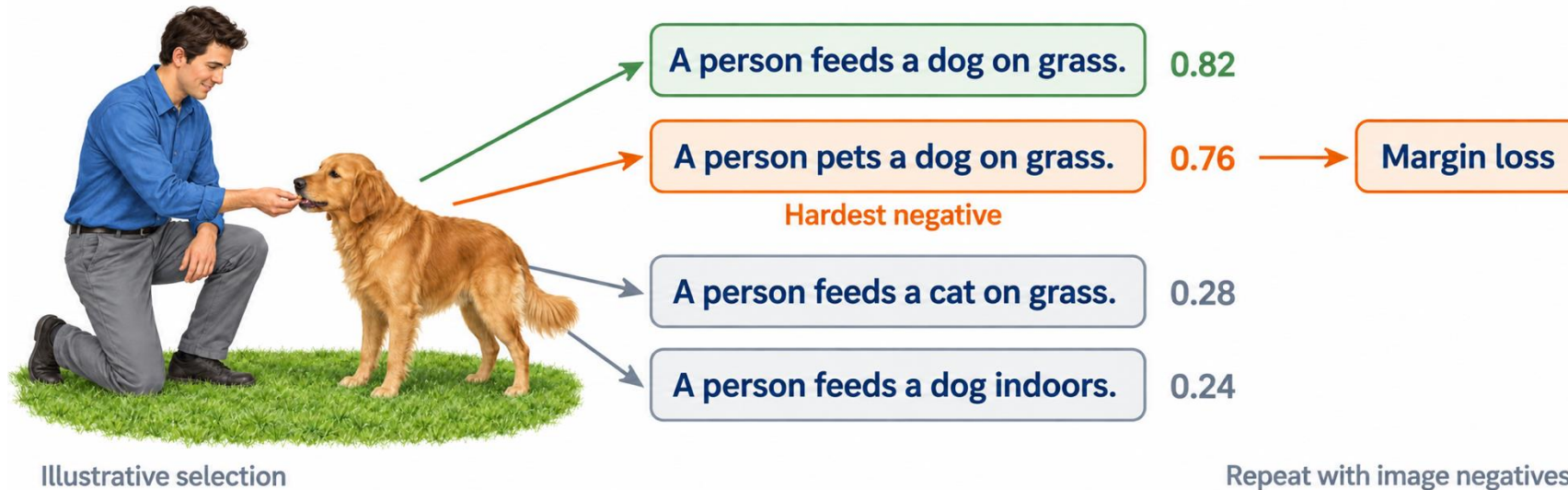


VSE++: Hard Negatives

$$T^* = \arg \max_{T' \neq T} s(I, T') \quad I^* = \arg \max_{I' \neq I} s(I', T)$$

$$L(I, T) = \max(0, \alpha - s(I, T) + s(I, T^*)) + \max(0, \alpha - s(I, T) + s(I^*, T))$$

Only the hardest violation contributes



VSE++: Retrieval Results

Configuration	Caption R@1	Image R@1
ResNet, 1K test	58.3	43.6
ResNet + fine-tuning, 1K	64.6	52.0
ResNet + fine-tuning, 5K	41.3	30.3

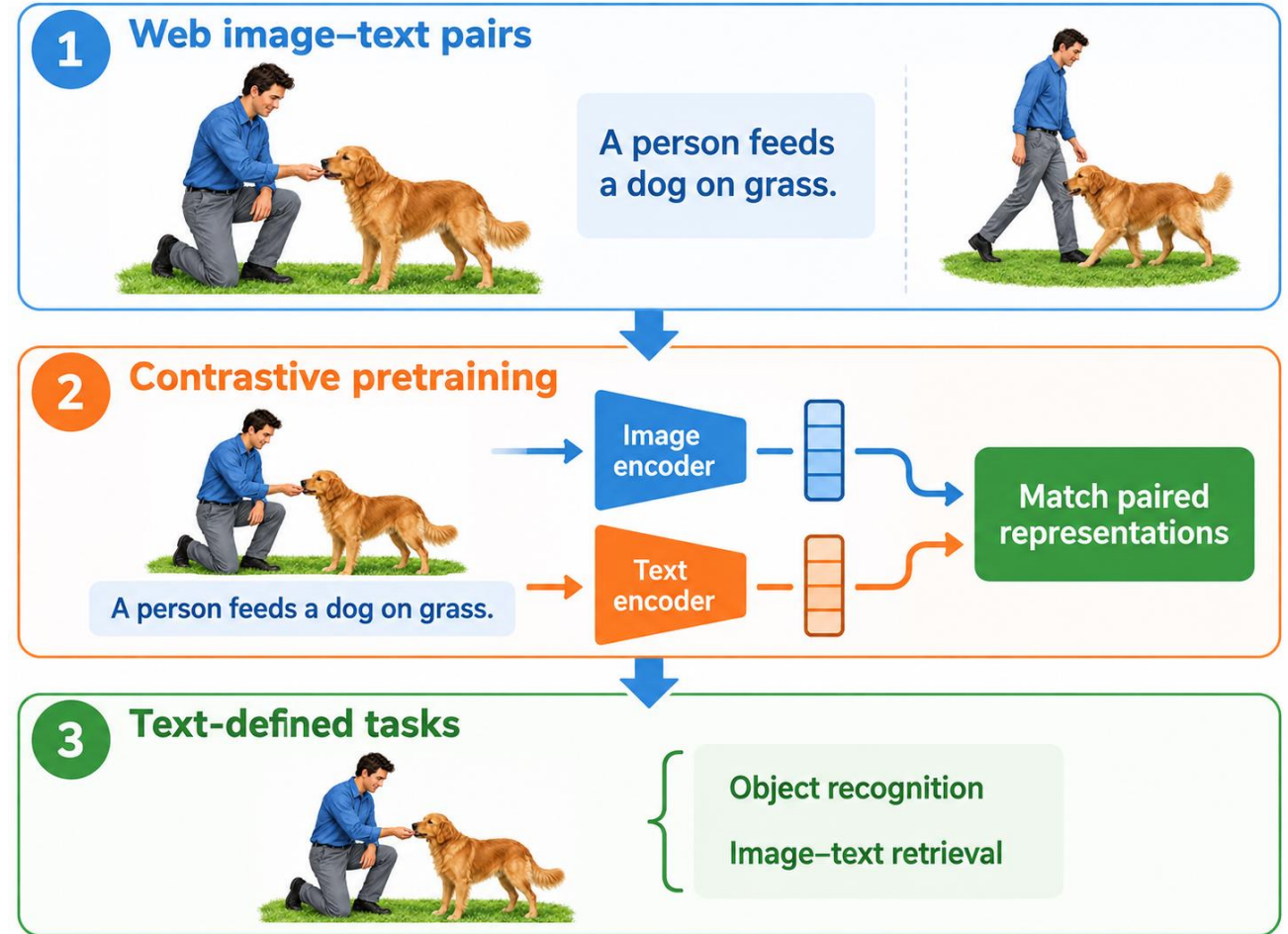
Candidate pool size changes the difficulty of retrieval

Scale Up: CLIP

Train on 400 million web image–text pairs: naturally occurring data!

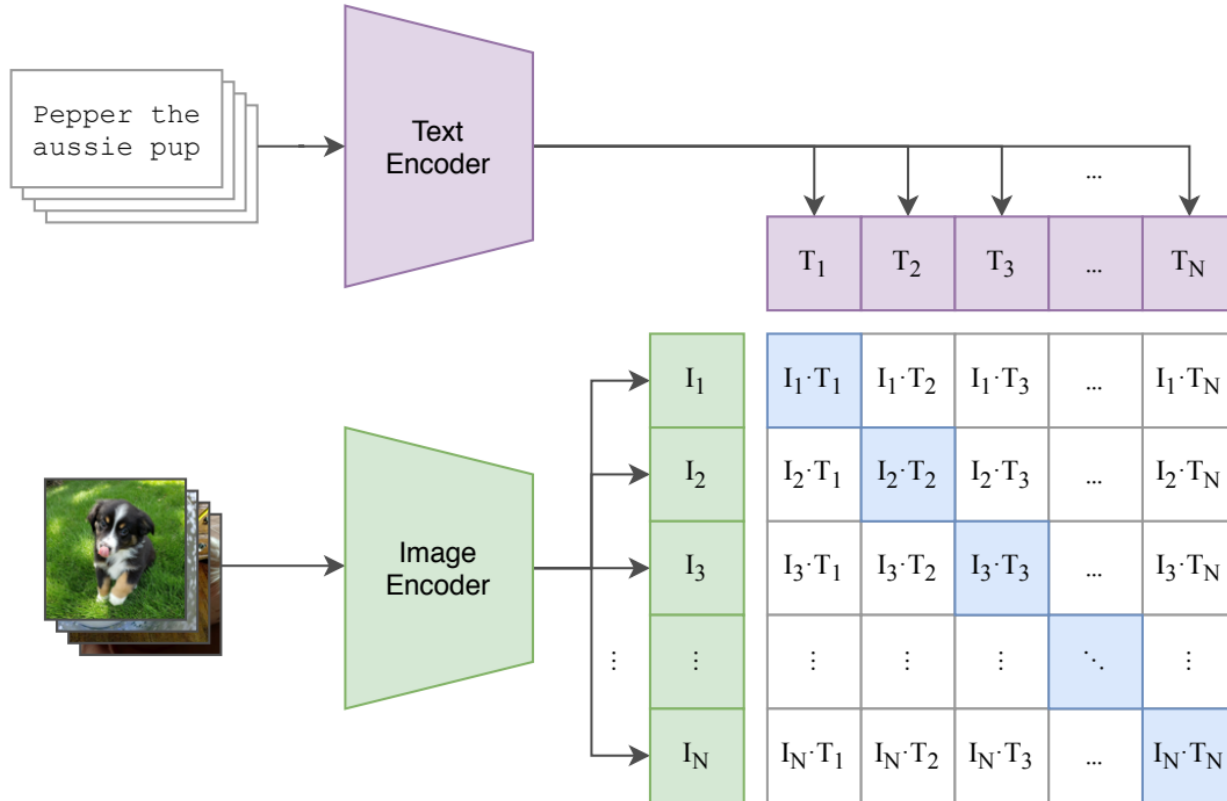
Learn separate encoders with a contrastive loss

Specify test-time categories through text



CLIP: Contrastive Pre-training

(1) Contrastive pre-training



One batch of N paired examples

$N \times N$ similarity matrix

Train both retrieval directions

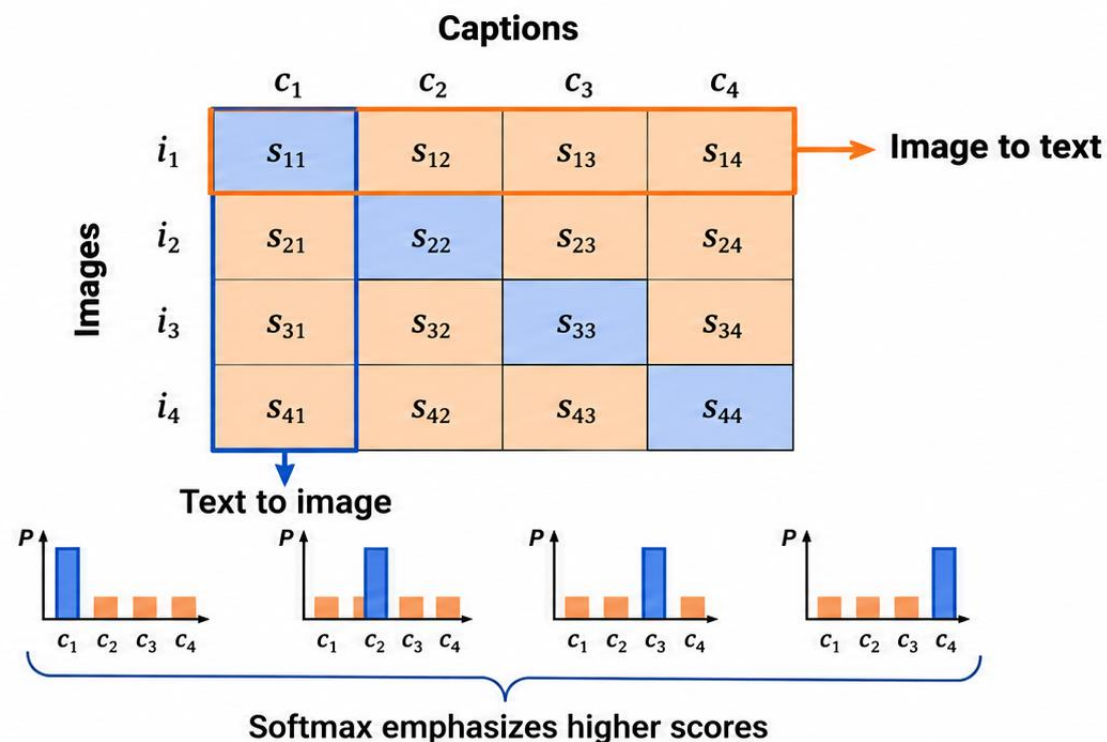
CLIP's Contrastive Objective

$$z_{ij} = \frac{v_i^T t_j}{\tau}$$

$$L_{i \rightarrow t} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_{ii})}{\sum_{j=1}^N \exp(z_{ij})}$$

$$L_{t \rightarrow i} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_{ii})}{\sum_{j=1}^N \exp(z_{ji})}$$

$$L = \frac{1}{2} (L_{i \rightarrow t} + L_{t \rightarrow i})$$



Harder negatives receive more weight through the softmax

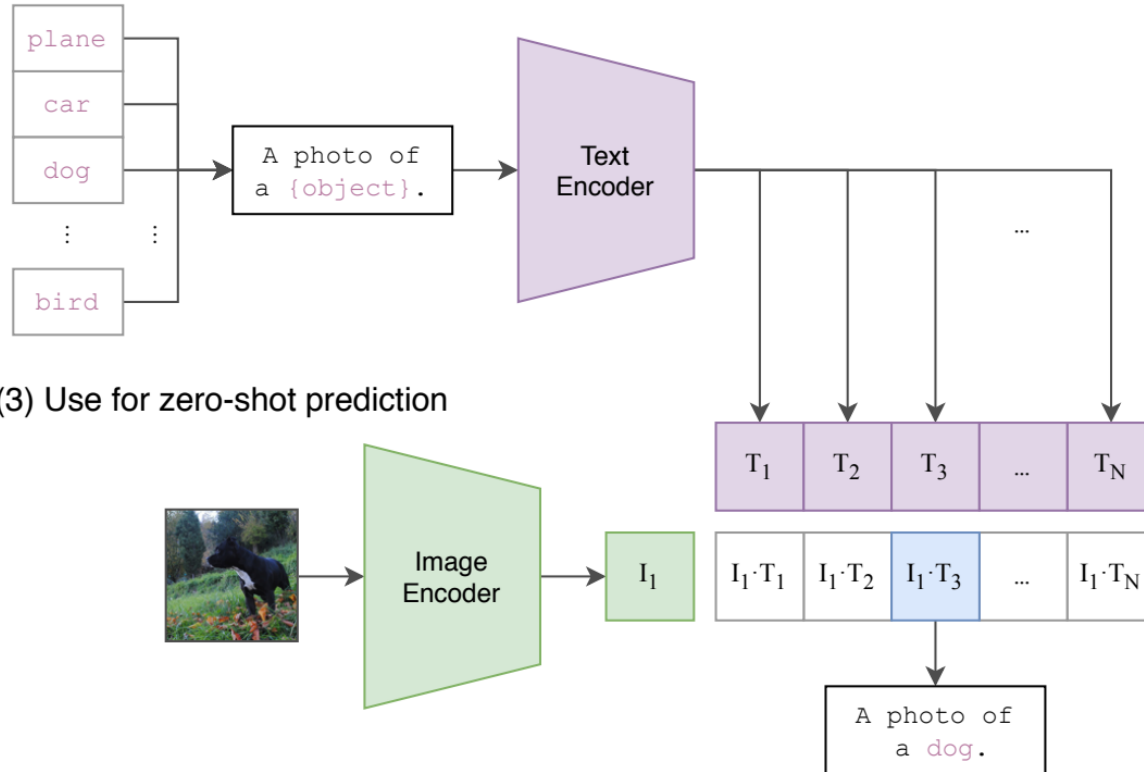
VSE++ and CLIP

Design choice	VSE++	CLIP
Loss	Hardest hinge violation	Symmetric cross-entropy
Negative Sampling	Maximum within batch	Softmax over batch
Training setting	COCO / Flickr30K	400M web pairs

Why do we choose the Softmax operation?
Check the **Bradley–Terry model**

CLIP: Zero-Shot Classification

(2) Create dataset classifier from label text



“a photo of a dog”

“a photo of a car”

Predict the text with the highest image similarity

CLIP: Results and Limits

Best reported zero-shot ImageNet accuracy: 76.2%

Fine-grained distinctions, counting and distribution shift remain hard

CLIP is the foundation model for a lot of follow-up vision-language papers (including VLMs); They spent a lot of effort cleaning the data

The CLIP Embedding Space Is Different from You Expected

Cosine still measures an angle

$$s = x^T y, \quad \|x\|_2 = \|y\|_2 = 1$$

How to interpret this score?

Reference model: independent, uniform unit vectors in d dimensions

This is a baseline for interpreting scores.



A human is feeding a dog on the grass

What's the Distribution of s ?

By rotational symmetry: $x = e_1$

$$s = x^T y = y_1$$

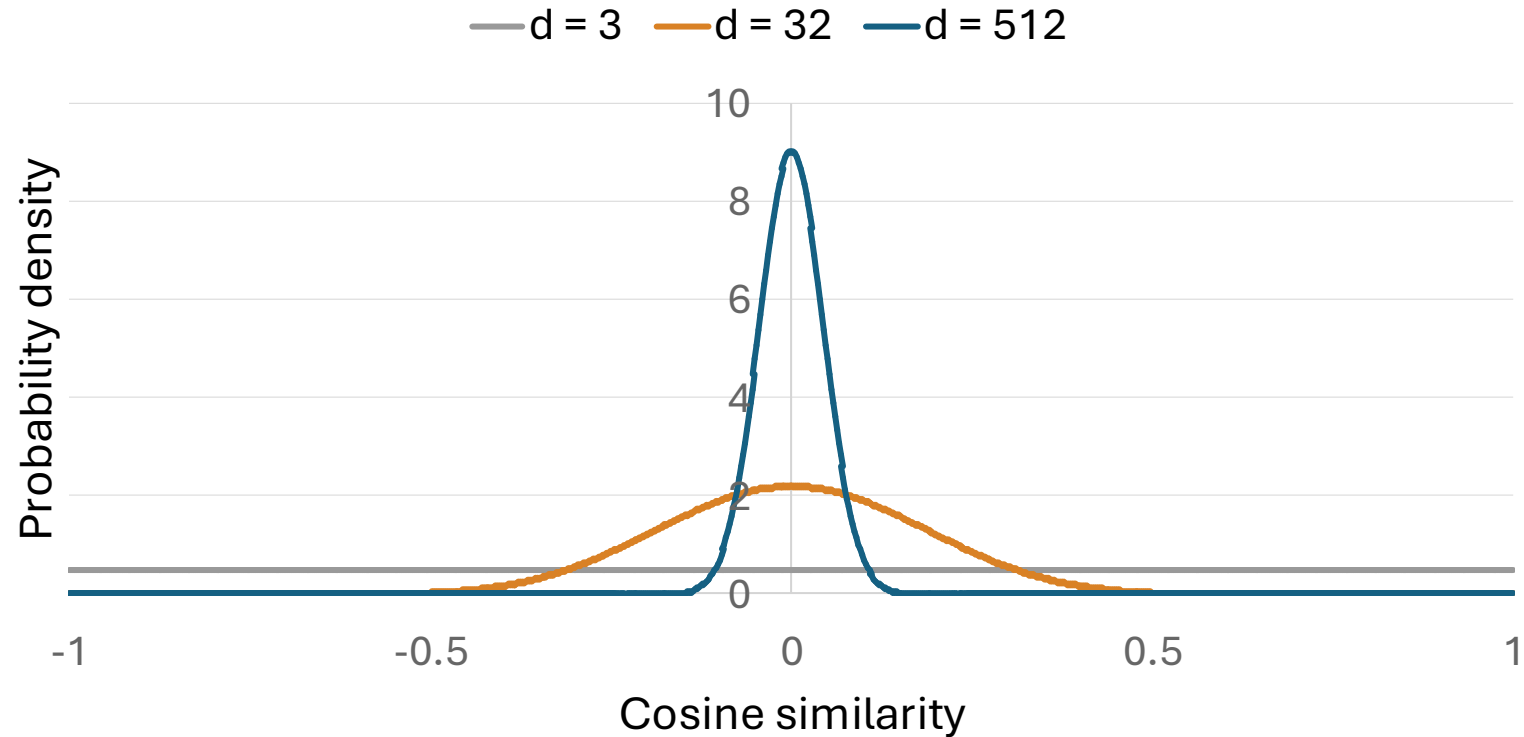
$$E[s] = 0$$

$$\sum_{k=1}^d y_k^2 = 1 \Rightarrow E[y_1^2] = \frac{1}{d}$$

$$\text{Var}(s) = \frac{1}{d} \quad \text{SD}(s) = \frac{1}{\sqrt{d}}$$

Dimension Controls the Cosine Distribution

Independent uniform unit vectors



Dimension d	SD of cosine
3	0.577
32	0.177
512	0.044

As d grows, random pairs concentrate near 90° .

Mind the Gap: CLIP's Embedding Space

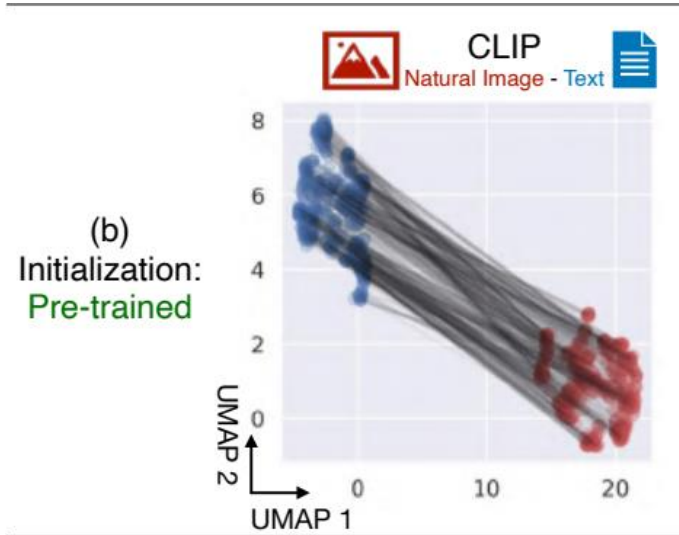
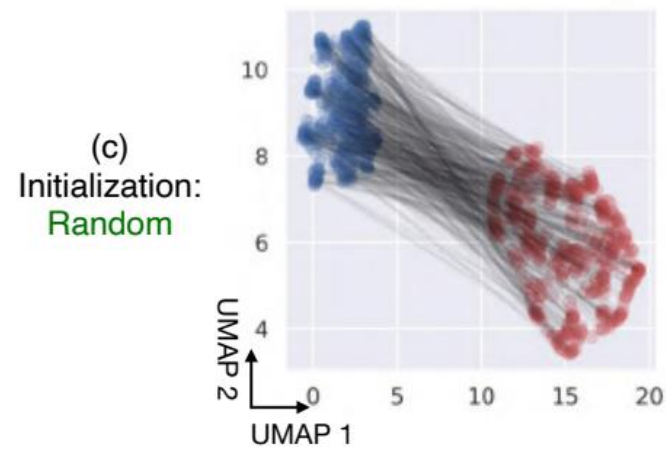


Image and text embeddings are in distinct regions

The separation is visible even at random init



A shared space does not require iid distributions

Modality Gap: Definition

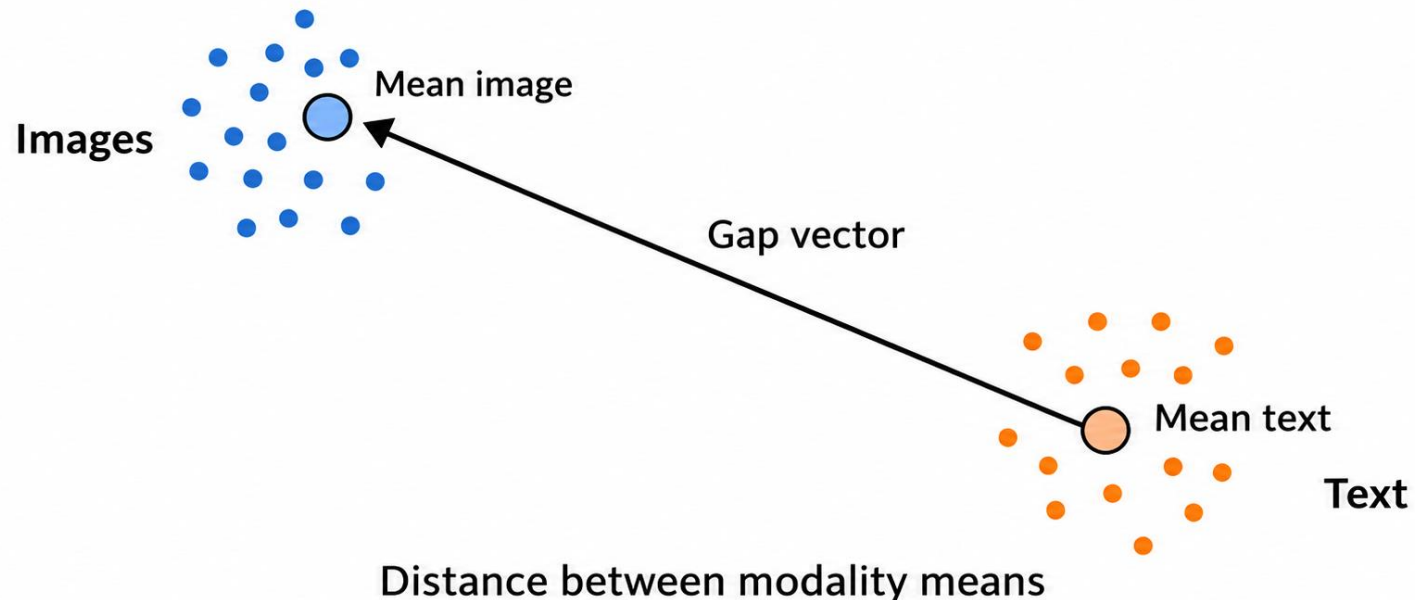
Let μ_i and μ_t be the mean normalized image and text vectors

$$\Delta = \mu_i - \mu_t$$

$$g = \|\Delta\|_2$$

This is only a measure for separation between modalities

The retrieval accuracy is still high



Modality Gap: Definition

Let μ_i and μ_t be the mean normalized image and text vectors

$$\Delta = \mu_i - \mu_t$$

$$g = \|\Delta\|_2$$

This is only a meas

The retrieval accur

Mind the Gap: Understanding the Modality Gap in Multi-modal Contrastive Representation Learning

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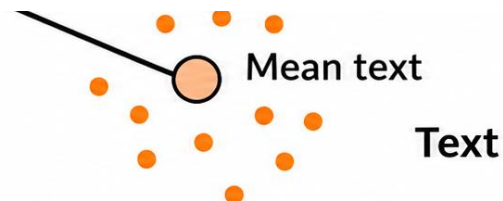
Columbia University
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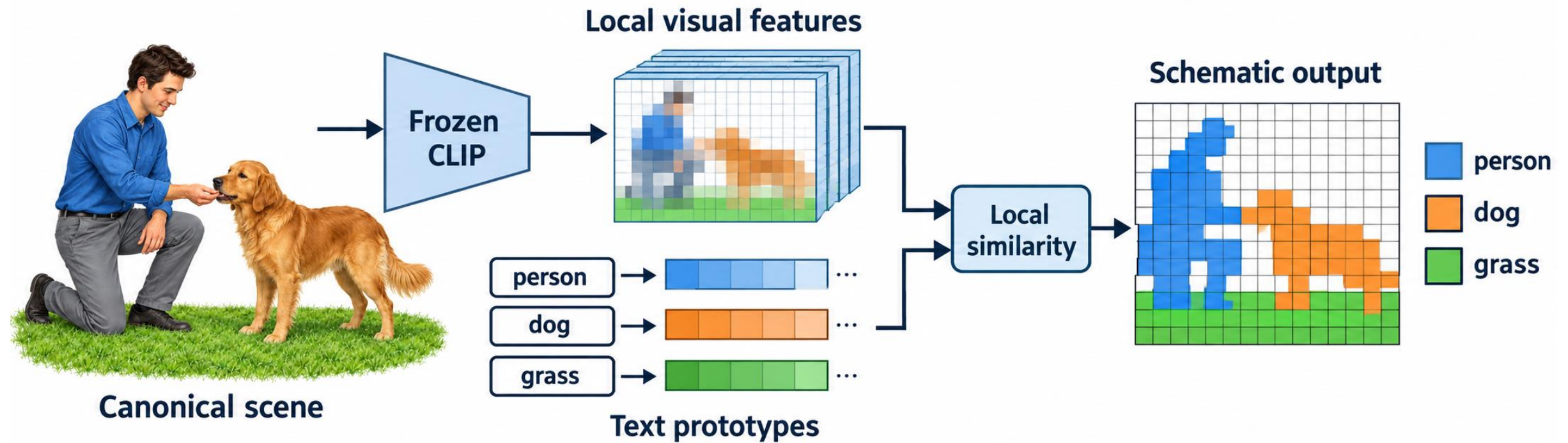
James Zou

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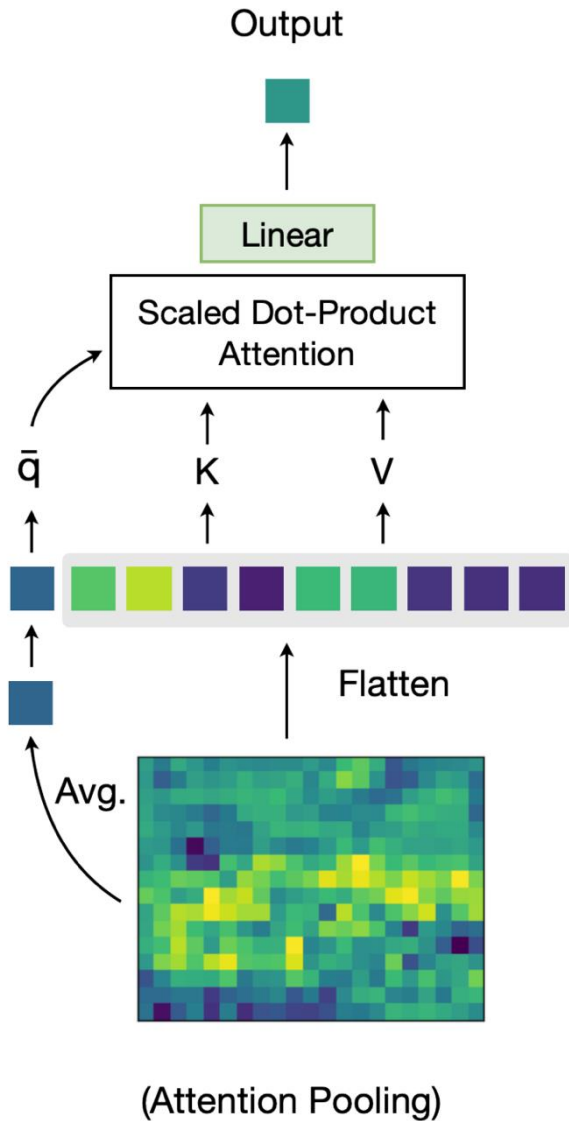
Distance between modality means

Use CLIP to get Pixel-Level Labels



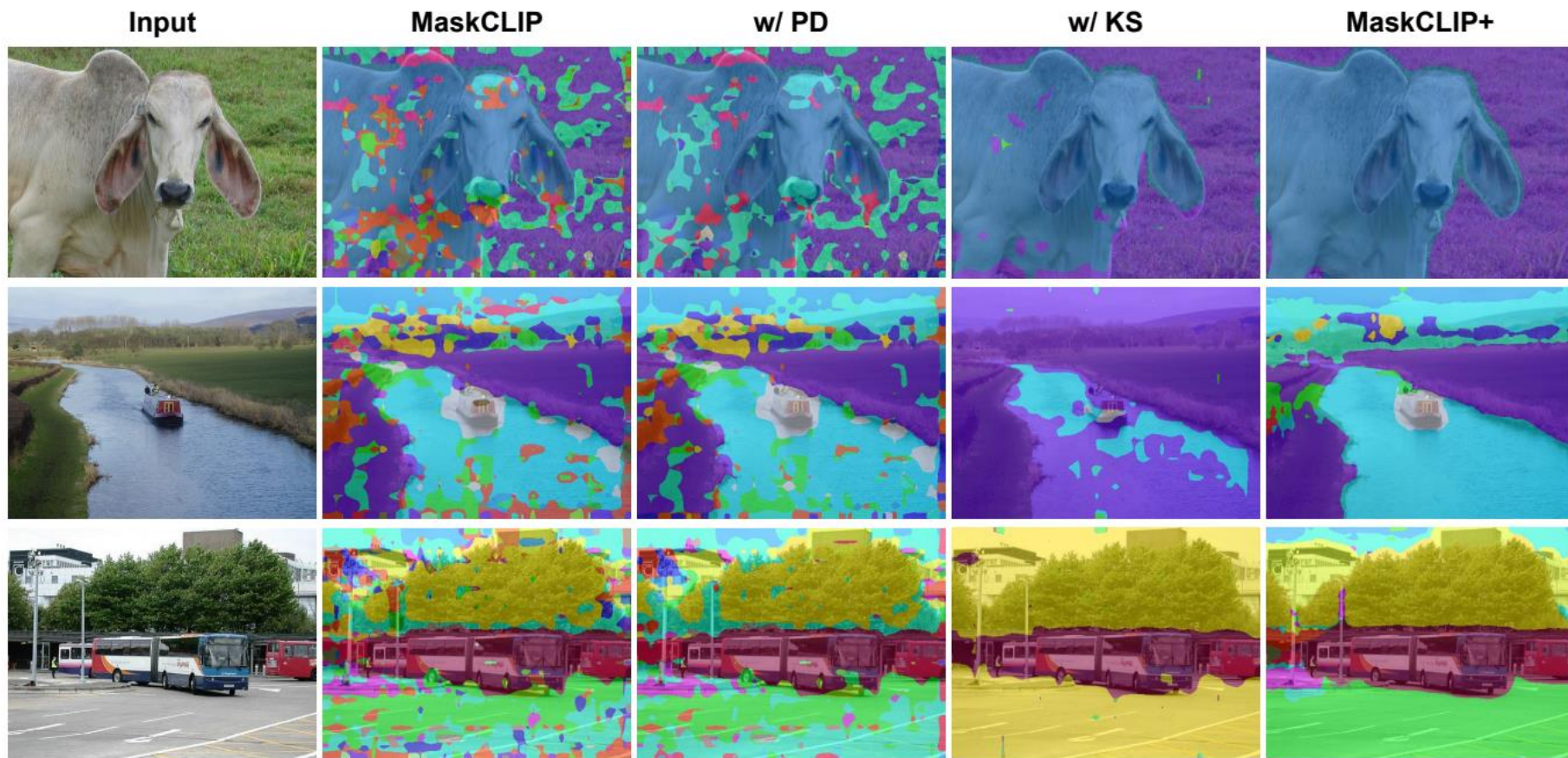
Frozen CLIP features + text prototypes → dense class scores

Use CLIP to get Pixel-Level Labels



- CLIP uses an “Attention Pooling” to aggregate information
- The output is a linear combination of $V(\text{feat})$ at each pixel
- So, we can compute
- $\text{sim}(V(\text{feat}_{i,j}), \text{text})$

MaskCLIP: Segmentation Examples



Local predictions are useful, but noisy; They proposed methods for “distillation”

MaskCLIP: Segmentation Examples

Extract Free Dense Labels from CLIP

Chong Zhou¹, Chen Change Loy¹, and Bo Dai^{2*}

¹ S-Lab, Nanyang Technological University

² Shanghai AI Laboratory

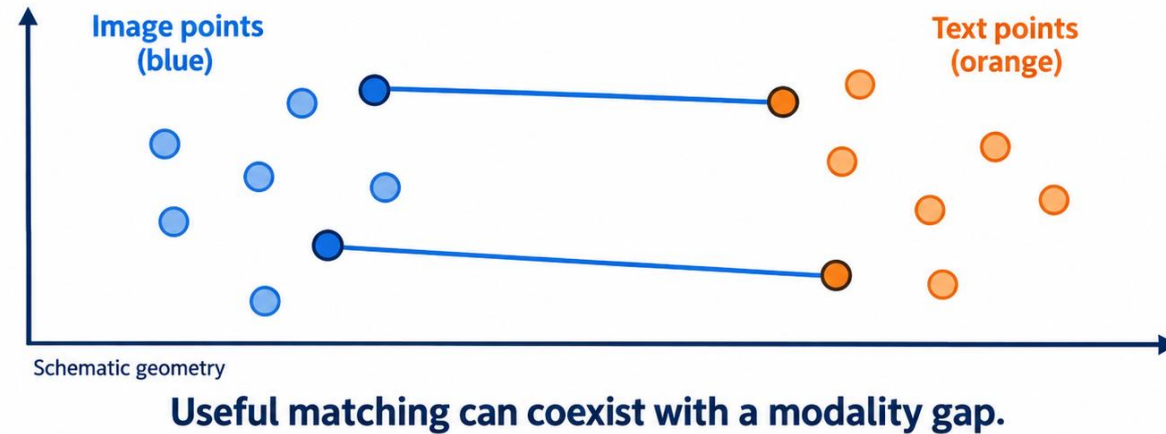
{chong033, ccloy}@ntu.edu.sg daibo@pjlab.org.cn

Local predictions are useful, but noisy; They proposed methods for “distillation”

Principles of Vision–Language Alignment

Common practice: Contrastive learning

Useful matching does not require identical modality distributions



Today's Agenda

Part 1: Principles for Vision-Language Alignment

- Contrastive learning and embedding geometry

Part 2: Principles for Vision-Language Alignment Structures

- Scene graphs and compositional structures

Caption-Level vs. Structured Alignment



Caption-level score

A person feeds a dog on grass.

Explicit components and roles

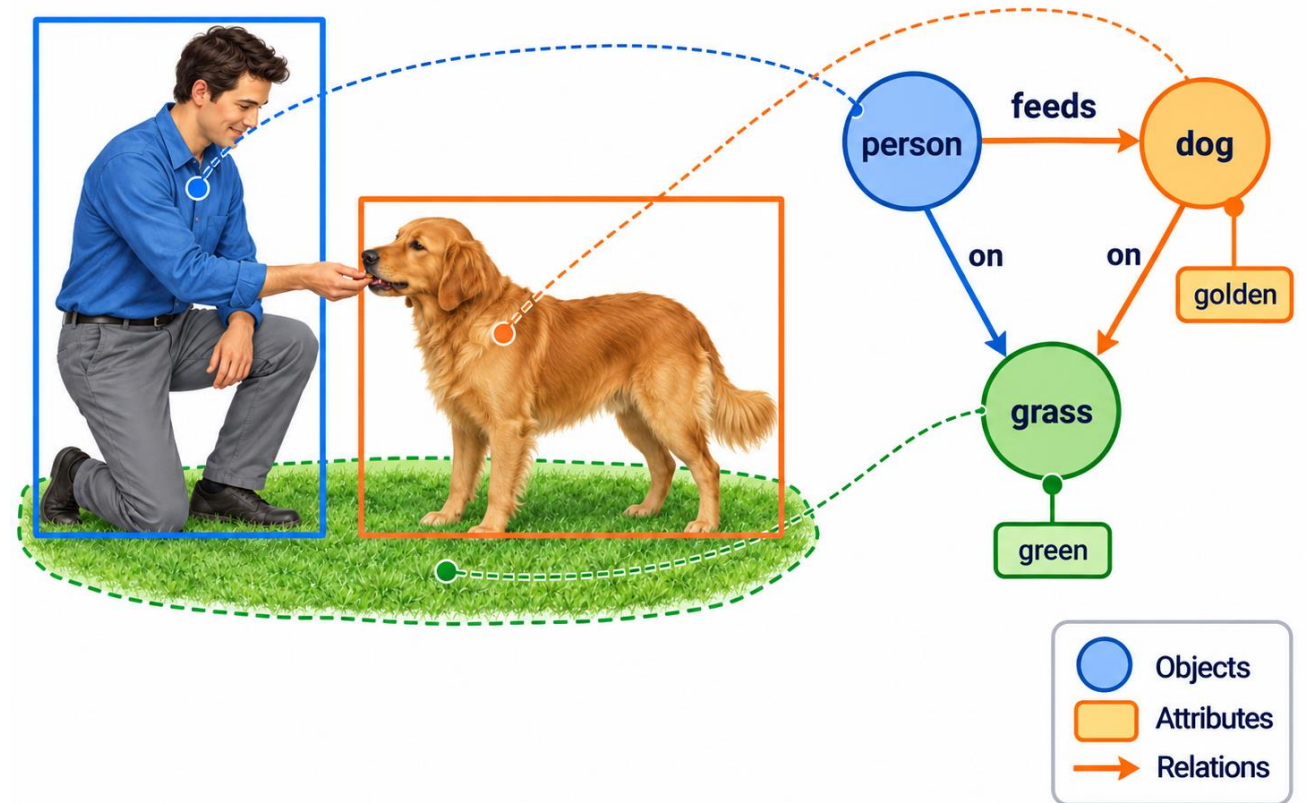


Scene Graphs: Explicit Visual Structure

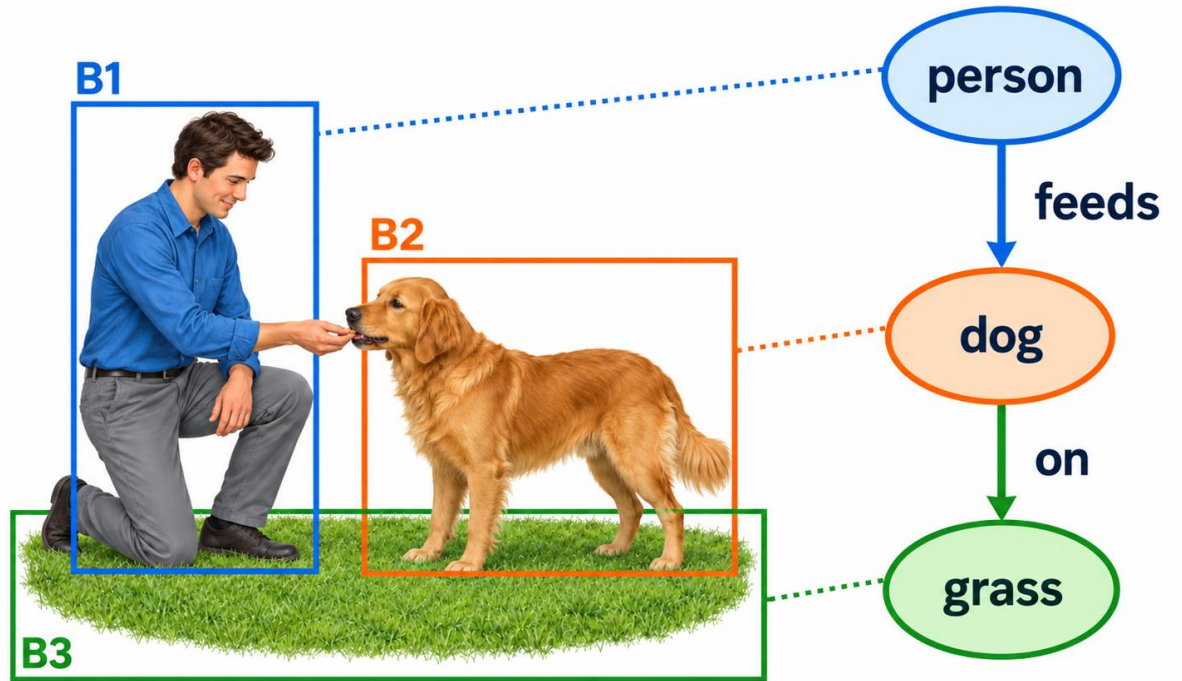
Objects identify entities

Attributes describe their properties

Directed relationships specify how entities interact



Scene Graphs and Grounding



**Grounding assigns
object nodes to regions**

Graph: objects, attributes,
relations

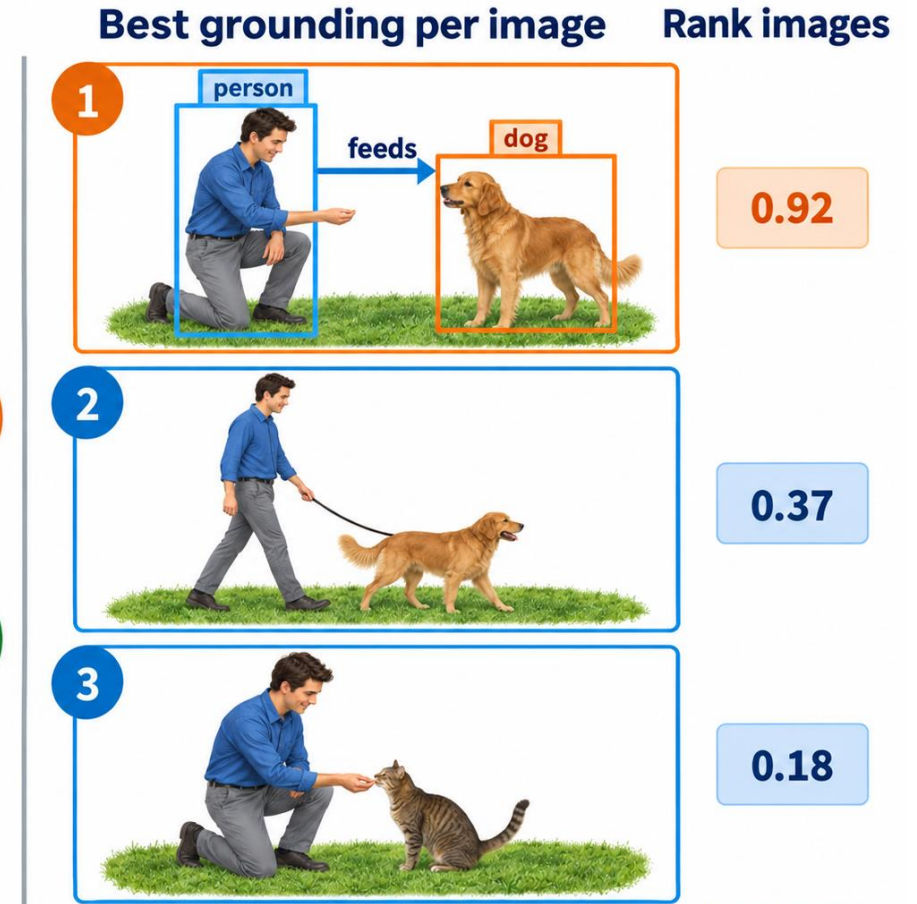
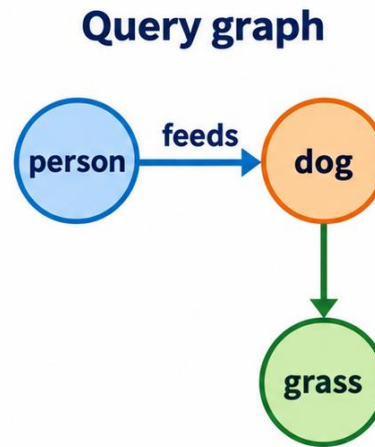
Grounding: assign each
object instance to an image
region

Image Retrieval by Graph Grounding

The query is a scene graph

Search each image for a compatible grounding

Rank images by their best grounding score



Illustrative ranking

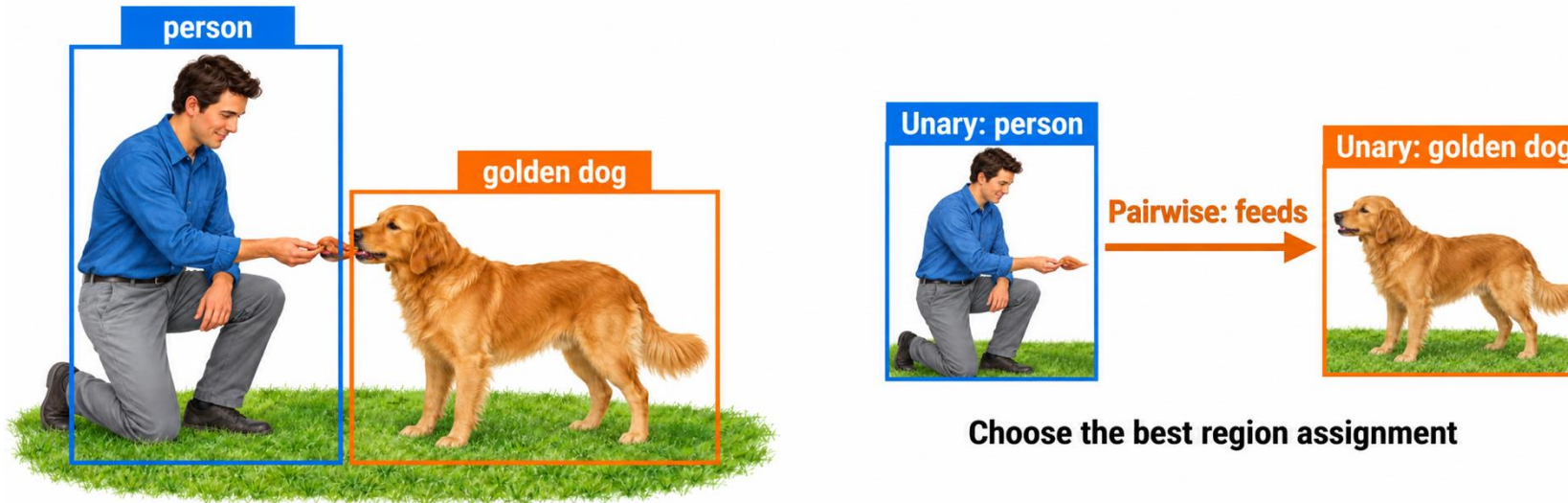
The Scene Graph Matching Model

$$\text{score}(\gamma) = \sum_o \log \varphi_o(\gamma_o) + \sum_{r=(o,o')} \log \psi_r(\gamma_o, \gamma_{o'})$$

Unary terms: does this region match the object and attributes?

Pairwise terms: do these regions fit the relationship?

Best grounding: choose γ with the highest score



Detailed Annotations Help, But They Are Expensive

Scene graph features	R@1 (%)	R@10 (%)	Median rank
Objects	11.3	34.7	28.0
Objects + attributes	12.7	42.0	17.5
Objects + attributes + relations	13.3	43.3	14.0

Are There Any Other Ways to Get These “Labels?”

VSE-C: Generating “Contrastive” Captions

Image



Original Caption

Three giraffes and a rhino graze from trees.

numeral noun indef. noun relation: graze from noun

Contrastive Adversarial Samples

noun:

Three cows and a rhino graze from trees.

numeral / indefinite article:

Three giraffes and three rhinos graze from trees.

relation:

Three giraffes and a rhino graze on trees.

Trees graze from three giraffes and a rhino.

VSE-C: Three Types of Caption Edits



A person feeds one dog
on the grass.

Noun



A person feeds one
cat on the grass.

Number



A person feeds
two dogs on the grass.

Relation



One dog feeds
a person on the grass.

These Contrastive Captions Are Hard to Distinguish

Image → caption R@1 (%) under this paper's protocol

Model	Clean pool	With attacks
VSE++	55.7	35.6

Winoground: Same Words, Different Meaning



Some plants surrounding a lightbulb



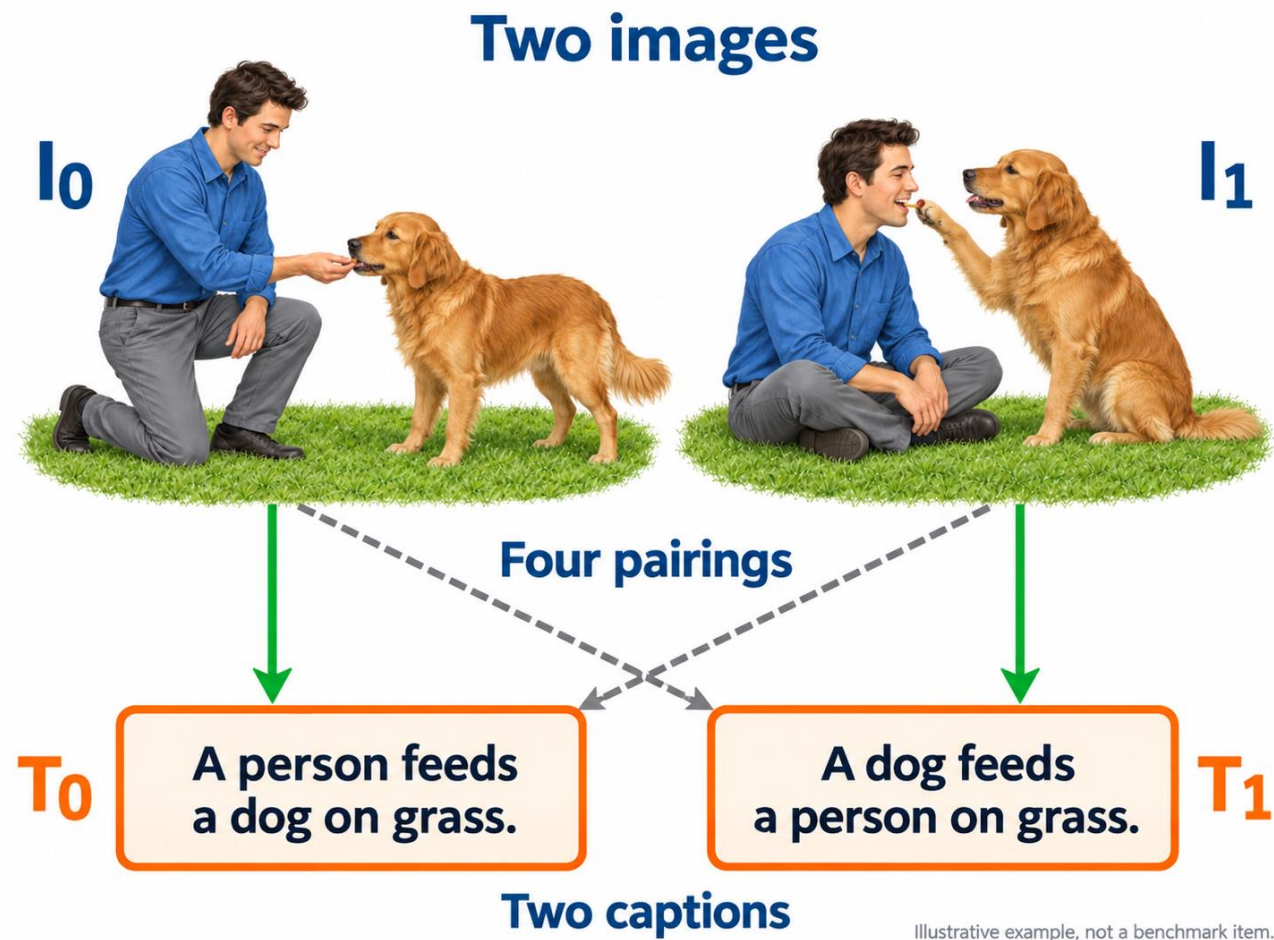
A lightbulb surrounding some plants

Winoground: Benchmark Design

400 examples with two images and two captions

Captions use the same words in a different order

Evaluate all four pairings



Illustrative example, not a benchmark item.

Winoground: Text, Image and Group Scores

$S_{ij} = s(I_i, T_j)$ (correct pairs on the diagonal)

Text: $S_{00} > S_{01}$ and $S_{11} > S_{10}$

Image: $S_{00} > S_{10}$ and $S_{11} > S_{01}$

Group: both text and image conditions hold

	T0	T1
I0	S00	S01
I1	S10	S11

Text:

compare within each row

S00 $\longleftrightarrow$ S01

S10 $\longleftrightarrow$ S11

Image:

compare within each column

S00

S01

$\updownarrow$

S10

$\updownarrow$

S11

Group:

satisfy both

S00 $\longleftrightarrow$ S01

$\updownarrow$

S10 $\longleftrightarrow$ S11

Winoground: Original Results

Model / baseline	Text (%)	Image (%)
VSE++ (COCO, ResNet)	22.75	8.00
CLIP (ViT-B/32)	30.75	10.50
Random scores	25.00	25.00
MTurk humans	89.50	88.50

Strong general transfer leaves a large compositional gap

VSE-C: Add Contrastive Captions to Training

VSE++: find a hard negative among other training pairs

VSE-C: also edit the positive caption itself

$$L = L_{\text{VSE++}} + \max_{T' \in C(T)} [\alpha + s(I, T') - s(I, T)]_+$$

The loss rewards the specific semantic distinction in the edit



These Contrastive Captions Are Hard to Distinguish

Image → caption R@1 (%) under this paper's protocol

Model	Clean pool	With attacks
VSE++	55.7	35.6
VSE-C (all edit types)	50.2	47.4

Generated negatives improve attack resistance, but lower clean R@1
Because the optimization of this model is still hard (at the time)

These Contrastive Captions Are Hard to Distinguish

Image → caption R@1 (%) under this paper's protocol

Learning Visually-Grounded Semantics from Contrastive Adversarial Samples

Haoyue Shi^{1*} **Jiayuan Mao**^{2*} **Tete Xiao**^{1*} **Yuning Jiang**³ **Jian Sun**³

¹: School of Electronics Engineering and Computer Science, Peking University, China

²: ITCS, Institute for Interdisciplinary Information Sciences, Tsinghua University, China

³: Megvii, Inc.

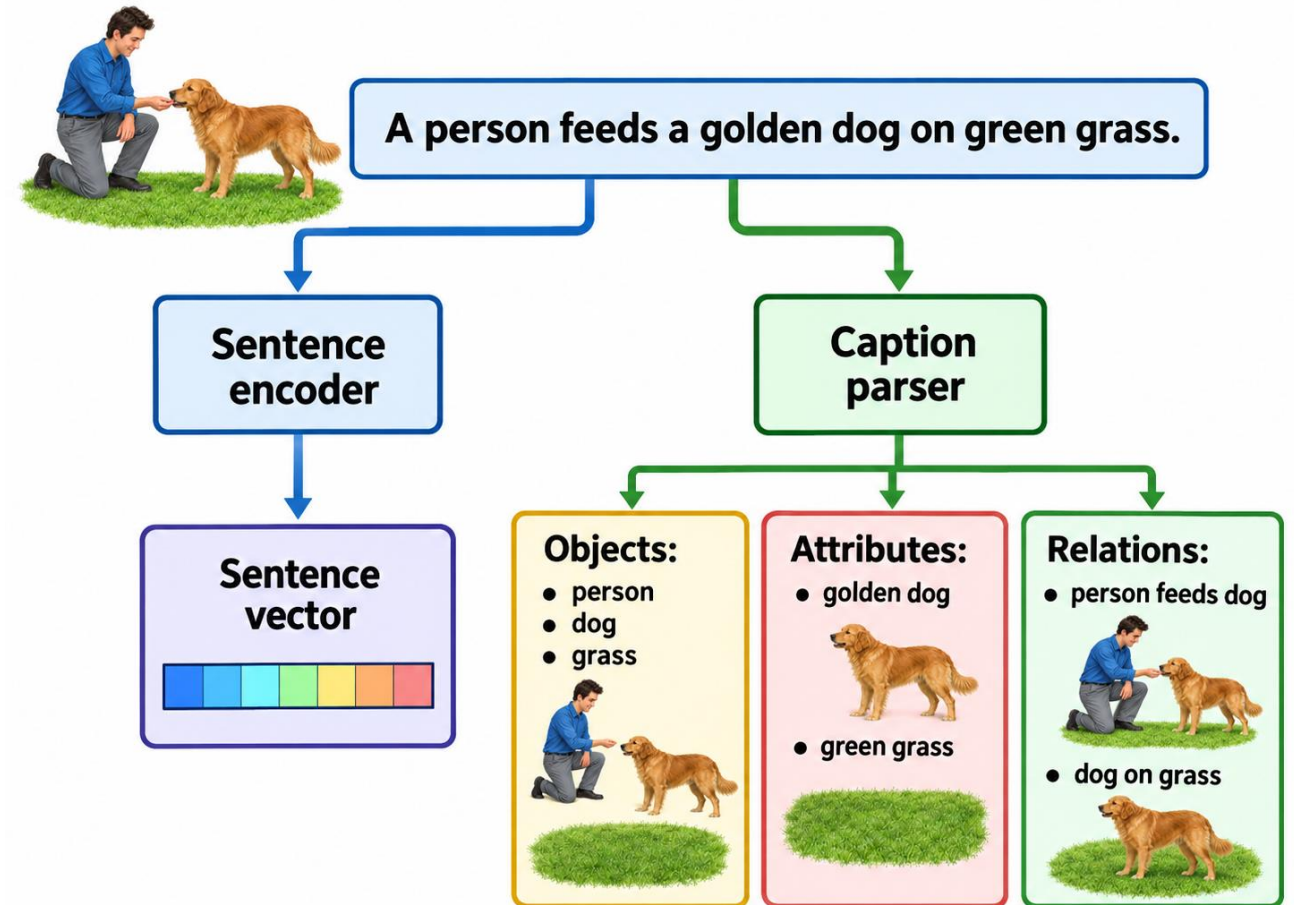
Generated negatives improve attack resistance, but lower clean R@1
Because the optimization of this model is still hard (at the time)

UniVSE: Factor the Meaning of a Caption

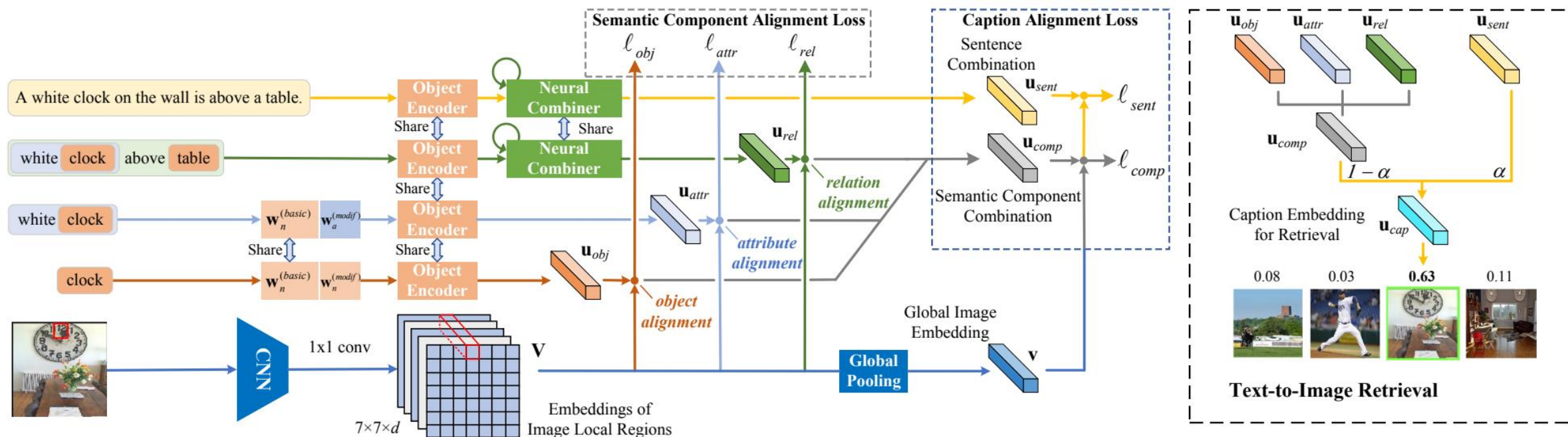
Parse objects, attributes and relations

Learn component embeddings

Retain a sentence representation alongside the components



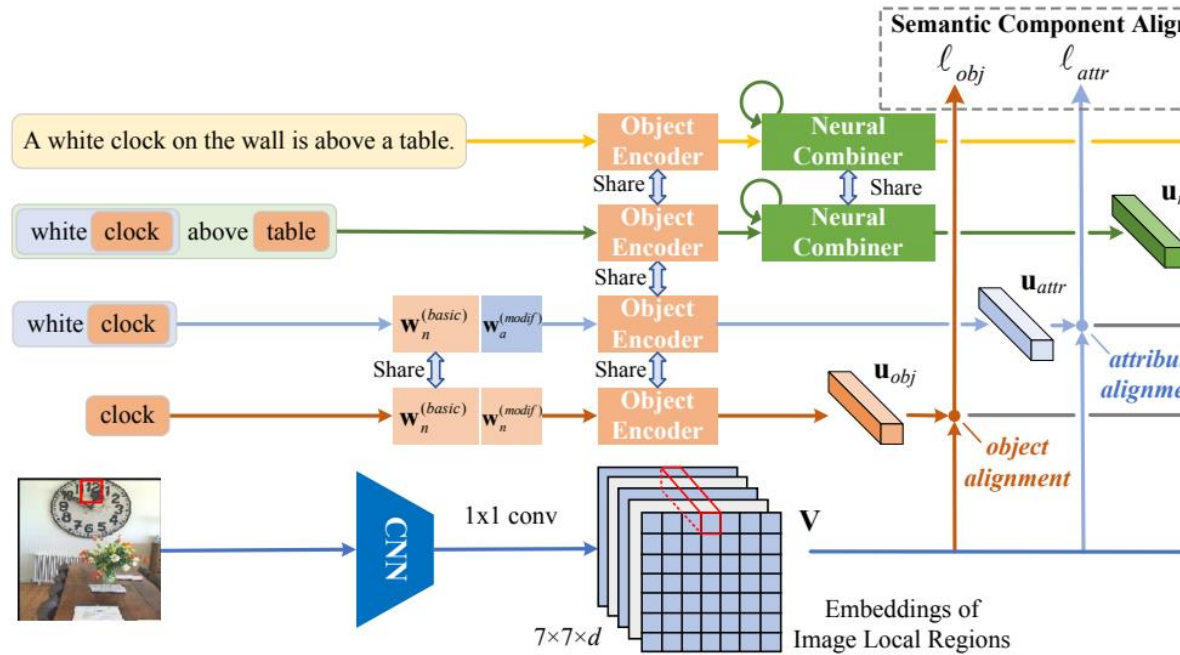
UniVSE: A Unified Embedding Model



Use MaskCLIP-like trick to align objects and attributes

Use Standard VSE++ to align relations and full sentences

UniVSE: A Unified Embedding Model



Objects and attributes: local



Region weights (schematic)

Relations and captions: global



person feeds dog

A person feeds a dog on grass.

UniVSE: relation aligns to whole image

Use MaskCLIP-like trick to align objects and attributes
Use Standard VSE++ to align relations and full sentences

UniVSE: Standard Retrieval

COCO 1K protocol • frozen visual encoder • R@1 (%)

Model	Caption retrieval	Image retrieval
VSE++	57.7	42.8
UniVSE	64.3	48.3

Structured training can also improve general retrieval

UniVSE: Coverage and Adversarial Retrieval

Image → caption R@1 (%) • 30,000-caption attack pools

Representation	Object	Attribute	Relation
VSE-C	41.1	26.7	35.5
UniVSE: sentence only	40.7	30.0	32.6
UniVSE: sentence + components	45.3	35.3	39.0

Structured training can also improve retrieval under hard example

UniVSE: Coverage and Adversarial Retrieval

Image $\rightarrow$ caption R@1 (%) • 30,000-caption attack pools

UniVSE: Robust Visual Semantic Embeddings via Structured Semantic Representations*

Hao Wu^{1,3,4,6,†,‡}, Jiayuan Mao^{5,6,†,‡}, Yufeng Zhang^{2,6,‡}, Yuning Jiang⁶, Lei Li⁶, Weiwei Sun^{1,3,4}, Wei-Ying Ma⁶

¹School of Computer Science, ²School of Economics, Fudan University

³Systems and Shanghai Key Laboratory of Data Science, Fudan University

⁴Shanghai Institute of Intelligent Electronics & Systems

⁵ITCS, Institute for Interdisciplinary Information Sciences, Tsinghua University, ⁶Bytedance AI Lab

Structured training can also improve retrieval under hard example

The Information-Theoretic Treatment of Alignment

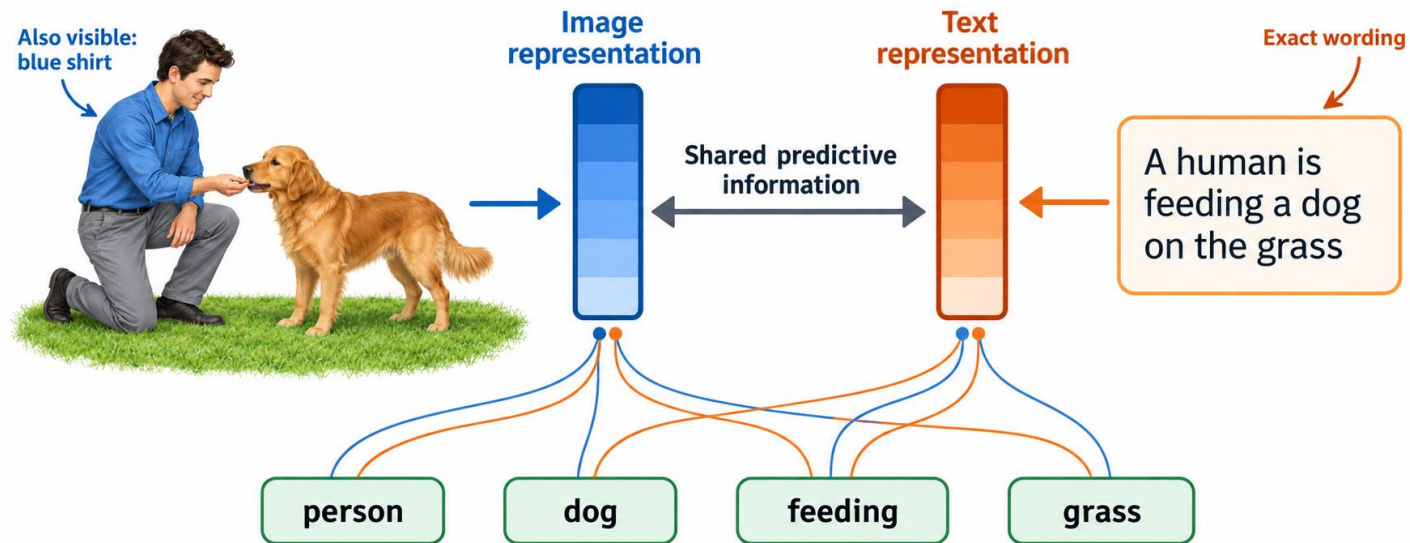
The Information-Theoretic Treatment of Alignment

Image and text embeddings: shared information across paired inputs.

$$I(Z_X; Z_T) \geq \log N - L_{\text{NCE}}$$

N: number of candidates

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{p(a,b)} \left[\log \frac{\exp(s(a,b))}{\sum_{\hat{b} \in \hat{B}} \exp(s(a, \hat{b}))} \right]$$



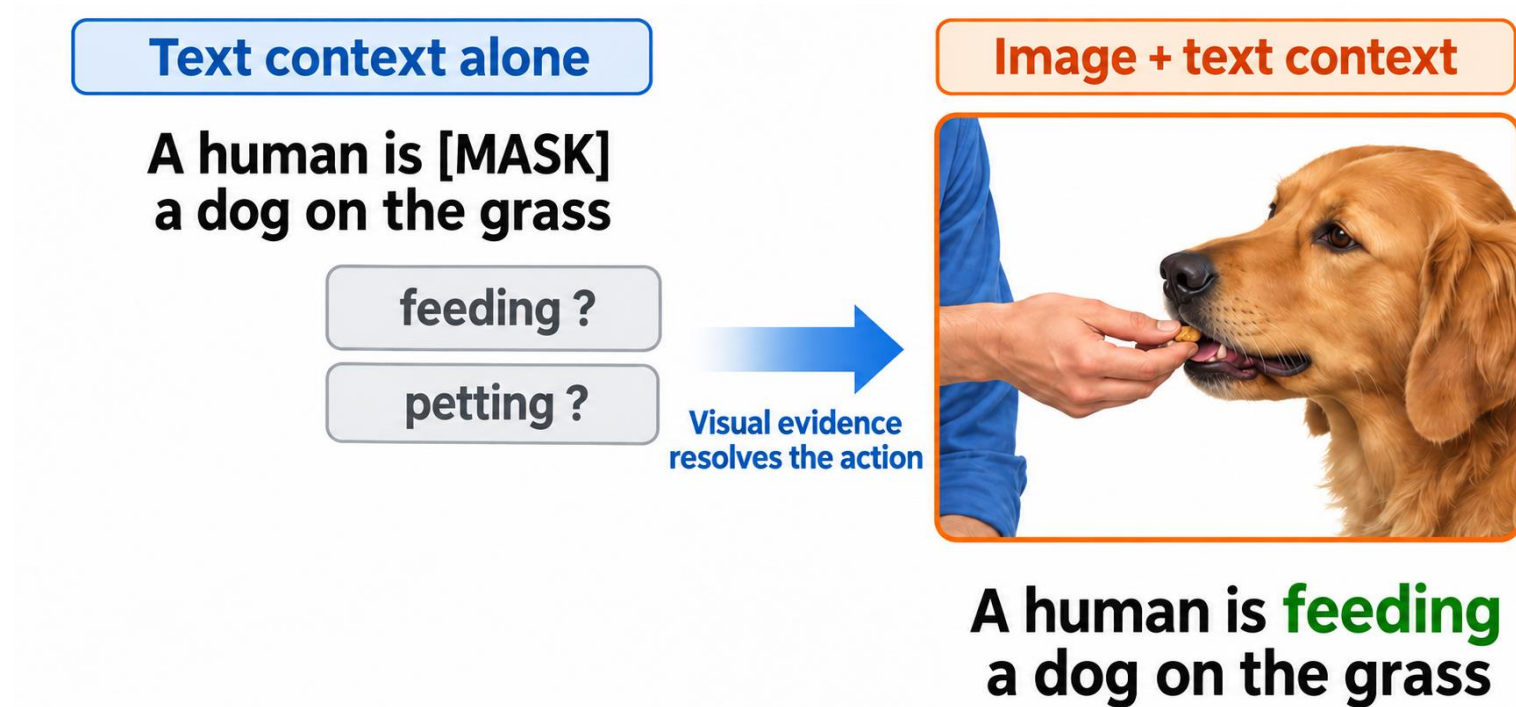
We are maximizing the mutual information between them!

The Information-Theoretic Treatment of Fusion

Y : masked word C : remaining caption X_I : image

$$H(Y | C) - H(Y | X_I, C) = I(Y; X_I | C)$$

Intuition: The image can reduce uncertainty that remains after reading the text.



Why Alignment and Fusion Work Together

Different views impose different prediction constraints.

ITC views: (I, T)

MLM views: $(y_{\text{mask}}, (I, \hat{T}))$

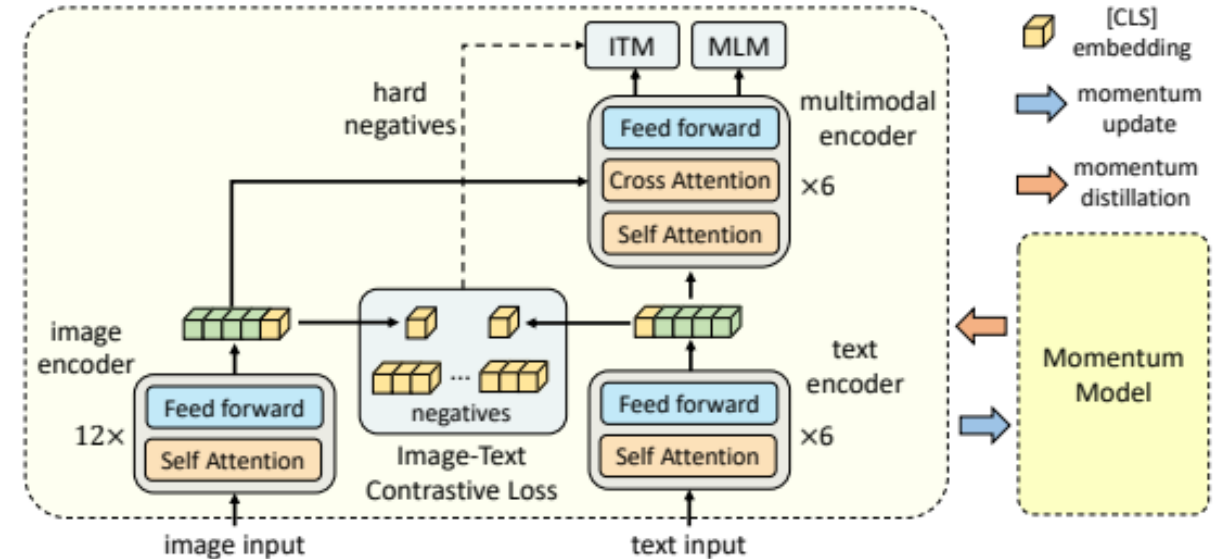
ALBEF: Alignment Before Fusion

ITC: match global image and text views

MLM: predict a word from image + text context

ITM: distinguish paired inputs from hard negatives

$$L = L_{ITC} + L_{MLM} + L_{ITM}$$



Principles of Vision–Language Alignment Structures

- Language has the explicit structure (relational)
- Modeling this structure is helpful:
 - Explicit scene graph-based parsing and retrieval
 - Generating training data (aka, augmentation: VSE-C)
 - Compositional models: align both primitives and full sentences (UniVSE)
- Autoregressive prediction is doing the job as well (due to the language structure)
- For feature learning, alignment and next-token-prediction focuses on different features

Question for Discussion

- VSE-C and Winoground suggest that these models learn biased image–text grounding—but is this necessarily a bad thing?
- What’s the advantage and disadvantage of “single embedding” representation for images and texts? (Be specific)
- What are naturally occurring (paired) data for different modalities in your domain?